Math 152 Lecture Notes: Week 3

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0.1 Summary of Last Week

Last week, we introduced the definite integral of a continuous function f over an interval $[a, b]$:

$$\int_a^b f(x)\mathrm{d}x$$

This was defined in terms of lower and upper approximations

$$L(f, \mathcal{P}) \leq \int_a^b f(x)\mathrm{d}x \leq U(f, \mathcal{P})$$

computed by partitioning the interval, multiplying the maximum and minimum values of the function on each part by the width of the part, and summing [draw picture]. This definition is motivated by the following interpretation: the integral of a nonnegative function $f \geq 0$ on $[a, b]$ is the area between the x axis and the graph of f. [Picture]

The Fundamental Theorem of Calculus connects differentiation to integration, stating that these processes are inverse to each other. Given a continuous function f, we can form a new function by integrating f:

$$F(x) = \int_a^x f(t)\mathrm{d}x$$

Theorem 1 (Fundamental Theorem of Calculus)**.** *Suppose that f is continuous.*

1. *The accumulation function F is an antiderivative for f, in the sense that $F'(x) = f(x)$.*
2. *If G is any antiderivative of f, then we can compute integrals of f using the formula*

$$\int_a^b f(x)\mathrm{d}x = G(b) - G(a)$$

In fact, any two antiderivatives for F differ by a constant. We introduced the indefinite integral

$$\int f(x)\mathrm{d}x$$

as a way to express all antiderivatives of f. By finding antiderivatives, we were able to compute integrals of polynomials, trigonometric functions, and other power functions.

0.2 Plan for this week

This week we will expand on each of the themes of the previous week: the integral as area, and computing integrals by relating them to differentiation. First, we will turn the chain rule for derivatives into the method of u-substitution for computing integrals. Then, we will see how to calculate the areas of more general regions in the plane using integrals.

1 Integration by Substitution

Recall that the chain rule tells us that

$$\frac{\mathrm{d}}{\mathrm{d}x}(g(f(x))) = g'(f(x))f'(x)$$

Read backwards, this tells us that if we recognize a function as having the form $h(f(x))f'(x)$, and g is an antiderivative for h, $h = g'$, then $g(f(x))$ is an antiderivative:

$$\int h(f(x))f'(x)\mathrm{d}x = g(f(x)) + C$$

Example 1. Consider the indefinite integral

$$\int \frac{\cos(x)}{\sin(x)^2}\mathrm{d}x$$

Taking $h(x) = x^{-2}$, $f(x)$, we have $f'(x) = \cos(x)$, so the function in the integral has the right form. The function $g(x) = -x^{-1}$ is an antiderivative of h, so our indefinite integral comes out to

$$g(f(x)) + C = \frac{-1}{\sin(x)} + C$$

Integration by u-substitution is a formalism for using this pattern to evaluate integrals, which can be applied to both indefinite and definite integrals. We will start with the indefinite integral case, and then see what we need to do to apply it to indefinite integrals.

We use the following notational device, which is a bit more flexible than directly matching the pattern $h(f(x))f'(x)$, but is mathematically equivalent: introduce a new variable $u = f(x)$, and rewrite

$$\int h(f(x))f'(x)\mathrm{d}x$$

as

$$\int h(u)\mathrm{d}u$$

using the formal expression $\mathrm{d}u = \frac{\mathrm{d}u}{\mathrm{d}x}\mathrm{d}x = f'(x)\mathrm{d}x$. Then if g is an antiderivative of h, this integral comes out to

$$g(u) + C = g(f(x)) + C$$

Example 2. Consider the integral

$$\int 2x\sqrt{1+x^2}\mathrm{d}x$$

Taking $u = 1 + x^2$, we have $\mathrm{d}u = 2x\mathrm{d}x$, and so our integral is

$$\int \sqrt{u}\mathrm{d}u = \frac{2}{3}u^{3/2} + C = \frac{2}{3}(1+x^2)^{3/2} + C$$

Remark. Using the letter u is just a convention, you can use any letter. Later when we do iterated substitution we will need to use other letters.

Remark. It takes a bit of practice to spot opportunities to apply substitution, and to find an expression for u that simplifies the integral. The homework will include a good number of examples of this for practice.

The u, $\mathrm{d}u$ notation makes it easy to deal with additional constants that come out when we take a derivative. For example, consider

$$\int \cos(5x)\mathrm{d}x$$

Taking $u = 5x$, we have $\mathrm{d}u = 5\mathrm{d}x$, which can be rewritten as $\mathrm{d}x = \frac{1}{5}\mathrm{d}u$. Thus the integral becomes

$$\int \frac{\cos(u)}{5}\mathrm{d}u = \frac{\sin(u)}{5} + C = \frac{\sin(5x)}{5} + C$$

In general, we have the following

Proposition. *If F is an antiderivative for f and k is a nonzero constant, then*

$$\int f(kx)\mathrm{d}x = \frac{1}{k}F(kx) + C$$

Proof. This follows from the same computation as above, taking $u = kx$, $\mathrm{d}u = k\mathrm{d}x$, $\mathrm{d}x = \frac{1}{k}\mathrm{d}u$, and so

$$\int f(kx)\mathrm{d}x = \int \frac{f(u)}{k}\mathrm{d}u = \frac{1}{k}F(u) + C = \frac{1}{k}F(kx) + C$$

Alternatively, you can just differentiate $\frac{1}{k}F(kx) + C$ and check that it is an antiderivative. □

Another case where we deal with constants like this is when the new u variable involves a power of x.

Example 3. Consider

$$\int \frac{x}{(1-x^2)^2}\mathrm{d}x$$

Taking $u = 1 - x^2$, we have $\mathrm{d}u = -2x\mathrm{d}x$, and so $x\mathrm{d}x = -\frac{1}{2}\mathrm{d}u$. Thus our integral can be rewritten

$$\int -\frac{1}{2}u^{-2}\mathrm{d}u = \frac{1}{2}u^{-1} + C = \frac{1}{2(1-x^2)} + C$$

Here is another straightforward consequence of the substitution method.

Proposition. *If F is an antiderivative of f and a is a constant, then*

$$\int f(x+a)\mathrm{d}x = F(x+a) + C$$

I leave it to you to check this, taking $u = x + a$, in which case $\mathrm{d}x = \mathrm{d}u$.

Example 4.

$$\int \sin(x - \pi/2)\mathrm{d}x = -\cos(x - \pi/2) + C$$

Sometimes we can do a bit more by expressing $\mathrm{d}x$ in terms of u and $\mathrm{d}u$, and using this to rewrite the integral.

Example 5. Consider

$$\int (1+\sqrt{x})^4 \mathrm{d}x$$

Taking $u = 1 + \sqrt{x}$, we have

$$\mathrm{d}u = \frac{1}{2\sqrt{x}}\mathrm{d}x = \frac{1}{2(u-1)}\mathrm{d}x$$

Thus we can write

$$\mathrm{d}x = 2(u-1)\mathrm{d}u$$

and so our integral becomes

$$\int 2u^4(u-1)\mathrm{d}u = 2\int [u^5 - u^4]\mathrm{d}u = 2(u^6/6 - u^5/5) + C = \frac{(1+\sqrt{x})^6}{3} - \frac{2(1+\sqrt{x})^5}{5} + C$$

It should be emphasized that all of these examples can be justified by the chain rule, and the calculations with $\mathrm{d}u$ and $\mathrm{d}x$ are a way to make this easier to apply. In other words, we have not given expressions like $\mathrm{d}x$ and $\mathrm{d}u$ a meaning, but we can use calculations with them to find the right instance of the chain rule to apply. (These expressions can be given a meaning, but this is outside the scope of this course.)

Another thing that the u, $\mathrm{d}u$ notation allows us to do is to apply multiple substitutions to an integral to simplify it in stages.

Example 6. Consider the integral

$$\int \sin(2x+1)(1+\cos(2x+1))^3 \mathrm{d}x$$

First, taking $u = 2x + 1$, we have $\mathrm{d}u = 2\mathrm{d}x$, and so the integral becomes

$$\frac{1}{2}\int \sin(u)(1+\cos(u))^3 \mathrm{d}u$$

Now we can apply substitution again, with a new variable $v = 1 + \cos(u)$. We then have $\mathrm{d}v = -\sin(u)\mathrm{d}u$, and so our integral becomes

$$-\frac{1}{2}\int v^3 \mathrm{d}v = -\frac{1}{8}v^4 + C = -\frac{1}{8}(1+\cos(u))^4 + C = -\frac{1}{8}(1+\cos(2x+1))^4 + C$$

1.1 Definite Integrals by Substitution

We now consider the application of substitution to compute *definite* integrals. One way to approach this is to use substitution to compute the indefinite integral, writing the result as a function of x and then using this to compute the definite integral using the fundamental theorem of calculus.

Example 7. To compute

$$\int_{-1}^{0} (2x+1)^2 \mathrm{d}x$$

we first compute the indefinite integral

$$\int (2x+1)^2 \mathrm{d}x$$

with the substitution $u = 2x + 1$, $\mathrm{d}u = 2\mathrm{d}x$

$$\rightsquigarrow \int \frac{1}{2}u^2 \mathrm{d}u = \frac{1}{6}u^3 + C = \frac{1}{6}(2x+1)^3 + C$$

Then we can use this antiderivative to evaluate the original definite integral:

$$\int_{-1}^{0} (2x+1)^2 \mathrm{d}x = \left[\frac{(2x+1)^3}{6}\right]\Bigg|_{-1}^{0} = \frac{1}{6} - \frac{-1}{6} = \frac{1}{3}$$

It turns out that there is a simpler way to do this. One way to see this is to go back to the justification for the method of u-substitution in terms of the chain rule. Remember that when we recognize a function as having the form $h(f(x))f'(x)\mathrm{d}x$, and g is an antiderivative of h, then $g(f(x))$ is an antiderivative of f. Therefore,

$$\int_a^b h(f(x))f'(x)\mathrm{d}x = g(f(b)) - g(f(a))$$

by the fundamental theorem of calculus. On the other hand, we can use g to compute definite integrals of h, and in particular we also have

$$g(f(b)) - g(f(a)) = \int_{f(a)}^{f(b)} h(u)\mathrm{d}u$$

where u is a new variable. Equating these expressions, we have the following.

Proposition (Change of Variable Formula)**.**

$$\int_a^b h(f(x))f'(x)\mathrm{d}x = \int_{f(a)}^{f(b)} h(u)\mathrm{d}u$$

As with indefinite integrals, the formalism of u-substitution makes this easier to apply. To illustrate this, we can consider the example above again.

Example 8. Again consider

$$\int_{-1}^{0} (2x+1)^2\mathrm{d}x$$

We will again use the substitution $u = 2x + 1$, $\mathrm{d}u = 2\mathrm{d}x$. To apply this to a definite integral, we replace the bounds of integration, -1 and 0, with the corresponding values of the new variable u:

$$\int_{u(-1)}^{u(0)} \frac{1}{2}u^2\mathrm{d}u = \int_{-1}^{1} \frac{1}{2}u^2\mathrm{d}u = \left[\frac{1}{6}u^3\right]\Big|_{-1}^{1} = \frac{1}{6} - \frac{-1}{6} = \frac{1}{3}$$

As in the indefinite integral case, we can apply this repeatedly to simplify an integral.

Example 9. Consider the integral

$$\int_0^{\pi/2} (\cos(2x))^2 \sin(2x)\mathrm{d}x$$

First we take $u = 2x$, $\mathrm{d}u = 2\mathrm{d}x$ and transform this to

$$\frac{1}{2}\int_0^{\pi} (\cos(u))^2 \sin(u)\mathrm{d}u$$

where the new bounds are $u(0) = 0$, $u(\pi/2) = \pi$. Now take $v = \cos(u)$, $\mathrm{d}v = -\sin(u)\mathrm{d}u$. The integral then becomes

$$-\frac{1}{2}\int_1^{-1} v^2\mathrm{d}v$$

with new bounds $v(0) = 1$, $v(\pi) = -1$. This is

$$\frac{1}{2}\int_{-1}^{1} v^2 \mathrm{d}v = \left[\frac{1}{6}v^3\right]\Big|_{-1}^{1} = \frac{1}{3}$$

Using this method, we can derive the following general formula.

Proposition. *For f continuous, k a nonzero constant*

$$\int_a^b f(kx)\mathrm{d}x = \frac{1}{k}\int_{ka}^{kb} f(u)\mathrm{d}u$$

This makes sense geometrically, because if $f \geq 0$, and R is the region below the graph of $f(kx)$, the integral on the right represents the area of a region obtain by horizontally stretching R by a factor of k. [Draw picture]

Similarly:

Proposition. *For f continuous, c a constant*

$$\int_a^b f(x+c)\mathrm{d}x = \int_{a+c}^{b+c} f(x)\mathrm{d}x$$

Geometrically, this corresponds to shifting the region horizontally by c. [Draw picture]

2 Areas of Plane Regions

So far, we have considered the area of regions between the x-axis and the graph of a function, which allows us to answer questions like the following.

Example 10. What is the area above the x-axis and below graph of $f(x) = 4 - x^2$? [Picture.] We first need to determine what the relevant interval is. The graph crosses the x-axis at the solutions to $4 - x^2 = 0$, which are -2 and 2. The function f is positive between -2 and 2, and positive outside this interval. Therefore, the area we want is

$$\int_{-2}^{2}(4 - x^2)\mathrm{d}x = [4x - x^3/3]\Big|_{-2}^{2} = (8 - 8/3) - (-8 + 8/3) = 32/3$$

2.1 The area between two curves

We can use a similar method to find the areas of more general plane regions. We first consider the case of two continuous functions f and g, where $0 \leq f \leq g$ on $[a, b]$. The area A above f and below g is the difference between the area below g and the area below f, and so

$$A = \int_a^b g(x)\mathrm{d}x - \int_a^b f(x)\mathrm{d}x.$$

[Draw picture] Using the linearity of the integral, this is equal to

$$\int_a^b [g(x) - f(x)]\mathrm{d}x$$

In fact, this formula for the area as an integral true as long as $f \leq g$, even if f and g take negative values. The point is that the formula does not change if we add a constant to f and g, because it depends only on the difference between g and f. Adding a constant corresponds to moving the region up or down, which does not change its area. Thus, the region might as well be above the x-axis.

Example 11. What is the area above $f(x) = -x$ and below $g(x) = 2 - 2x - x^2$ [Draw picture] As before, we need to find the relevant interval. The graphs cross at solutions to $f(x) = g(x)$, that is $-x = 2 - 2x - x^2 \rightsquigarrow x^2 + x - 2 = 0 \rightsquigarrow (x+2)(x-1) = 0$, i.e -2 and 1. You can check by plugging in points that $f(x) \leq g(x)$ on $[-2, 1]$, and $f(x) > g(x)$ outside this interval. The area we want is then

$$\int_{-2}^{1} [g(x) - f(x)]\mathrm{d}x = \int_{-2}^{1} [2 - x - x^2]\mathrm{d}x$$

$$= [2x - x^2/2 - x^3/3]\Big|_{-2}^{1} = (2 - 1/2 - 1/3) - (-4 - 2 + 8/3) = 7/6 - (-10/3) = 27/6 = 9/2$$

There is another way to explain why the formula for the area of the region between f and g is

$$\int_{a}^{b} [g(x) - f(x)]\mathrm{d}x.$$

The function $h(x) = g(x) - f(x)$ represents the cross-sectional height of the region at x [draw picture].

To help us think about the integral of this function, I will introduce another point of view of the definition of the integral of a function f. Given a partition $\mathcal{P} = \{x_0 < x_1 < \cdots < x_{n-1} < x_n\}$, and a choice of point $t_i \in [x_{i-1}, x_i]$, the sum

$$\sum_{i=1}^{n} f(t_i)\Delta x_i$$

is called a *Riemann sum*. If R is any Riemann sum, we have

$$L(f, \mathcal{P}) \leq R \leq U(f, \mathcal{P})$$

Thus if $\mathcal{P}$ is sufficiently fine so that $L(f, \mathcal{P})$ and $U(f, \mathcal{P})$ are very close together, any Riemann sum is close to the true value of the integral. In fact, the integral can be defined as a kind of limit of Riemann sums.

Returning to the integral of the height function h, we can approximate the area of the region with rectangles of with height given by h, the sum of the areas of these rectangles also approximates the integral of h, and the approximation can be made arbitrarily close by taking a sufficiently fine partition. [Draw picture.]

2.2 Computing area with horizontal cross sections

The same argument as for vertical cross sections also words for horizontal cross sections. [Draw picture]. If $w(y)$ represents the width of the cross section of a region between $y = a$ and $y = b$ at height y, then the area can be expressed as the integral

$$A = \int_{a}^{b} w(y)\mathrm{d}y$$

If the region lies between curves $x = f(y)$ and $x = g(y)$, where $f \leq g$, then $w(y) = g(y) - f(y)$.

Example 12. Consider the curves $x = y^2$ and $x = 8 - y^2$. [Draw picture] These intersect at $(4, 2)$ and $(4, -2)$, and enclose a region with are

$$A = \int_{-2}^{2} ((8 - y^2) - y^2)\mathrm{d}y = \int_{-2}^{2} [8 - 2y^2]\mathrm{d}y = \left[8y - \frac{2}{3}y^3\right]\Bigg|_{-2}^{2} = \frac{64}{3}$$

For a general curve forming the boundary of our region, we may have to solve for x as a function of y (or vice versa) to set up the integral. There may be multiple solutions to this equation, in which case we choose the solution based on geometric considerations. Let me illustrate this with an example

Example 13. Find the area enclosed by $y = 1$, $y = 2$ and $y = x^2 - 1$. [Draw picture] Here y ranges from 1 to 2, and the left and right curves are obtained by solving for x in $y = x^2 - 1$, yielding two solutions $x = \sqrt{1+y}$ and $x = -\sqrt{1+y}$. Looking at the picture, the first of these is the right boundary and the second the left, so the integral we want is

$$\int_1^2 [\sqrt{1+y} - (-\sqrt{1+y})]\mathrm{d}y = \int_1^2 2\sqrt{1+y}\mathrm{d}y = \int_2^3 2\sqrt{u}\mathrm{d}u = \left[\frac{2}{3}u^{3/2}\right]\Big|_2^3 = \frac{2}{3}(3^{3/2} - 2^{3/2})$$

In general, we can either compute the area as an integral in x or an integral in y. One of these approaches may be more convenient than the other. Here is an example where both are feasible.

Example 14. Consider the area between $y = -2x$ and $y = x^2$. [Draw picture]. These curves intersect at $(0, 0)$ and $(-2, 4)$. Let's first evaluate the area as an integral in x. Here the upper curve is $y = -2x$ and the lower $y = x^2$, so the cross sectional height is $h(x) = -2x - x^2$ and the area is

$$A = \int_{-2}^0 [-2x - x^2]\mathrm{d}x = \left[-x^2 - \frac{x^3}{3}\right]\Big|_{-2}^0 = 0 - (-4 + 8/3) = 4/3$$

To evaluate this as an integral in y, we need to express x as a function of y to obtain expressions for the right and left curves. Here the right curve is $y = -2x$, for which we have $x = -y/2$. The left curve is $y = x^2$, for which we have $x = -\sqrt{y}$ in this region. The area is then

$$A = \int_0^4 [-y/2 + \sqrt{y}]\mathrm{d}y = \left[\frac{2}{3}y^{3/2} - y^2/4\right]\Big|_0^4 = \frac{2}{3}8 - 4 = 4/3$$

2.3 Decomposing a region into simpler pieces

Sometimes we may not be able to cleanly write the area as a single integral in either x or y, because the top, bottom, left or right curves are composed of segments defined by different equations. In situations like this, a good approach is to decompose the region into smaller regions whose area can be computed with a simple integral, and then to add the resulting areas together. I will illustrate how to do this with a few examples.

Example 15. Consider the region enclosed by the curves

$$y = 2 - x^3$$
$$y = x + 2$$
$$y = -x - 4$$

[Draw picture] Whether we consider horizontal or vertical cross sections, we see that one of the boundary curves has two distinct segments defined by different equations. Seeking to evaluate the area as an integral in x, we can divide the region into two subregions, to the left and right of $x = 0$, where the two segments of the upper boundary meet. [Draw picture]. The region to the left of 0 is a triangle; we could evaluate its area as an integral, but we may as well just use the standard formula for the area of a triangle. In this case, the base is 6 and the height is 3, so the area is 9. The region to the right of $x = 0$ is bounded below $y = -x - 4$ and above by $y = 2 - x^3$, with x ranging from 0 to 2. Thus the area of this region is

$$\int_0^2 [2 - x^3 - (-x - 4)]\mathrm{d}x = \int_0^2 [6 + x - x^3]\mathrm{d}x = \left[6x + \frac{x^2}{2} - \frac{x^4}{4}\right]\Big|_0^2 = 12 + 2 - 4 = 10$$

Thus the total area of the region is 19.

Example 16. Consider the region bounded by the curves

$$y = 1 - x^2$$

$$y = -1$$

$$y = x - 1$$

[Draw picture.] Here we can divide this into two regions: one above the x-axis and one below. For the area above the x-axis, we can compute it as an integral in x:

$$\int_{-1}^{1} [1 - x^2] \mathrm{d}x = \left[x - \frac{x^3}{3} \right] \Big|_{-1}^{1} = 4/3$$

For the area below the x-axis, we can compute it as an integral in y. The left curve is given by $x = -\sqrt{1-y}$, and the right by $x = y + 1$, so the area is

$$\int_{-1}^{0} [y + 1 + \sqrt{1-y}] \mathrm{d}y$$

To compute this, we make the substitution $u = 1 - y$, transforming the integral into

$$-\int_{2}^{1} [2 - u + \sqrt{u}] \mathrm{d}u = \int_{1}^{2} [2 - u + \sqrt{u}] \mathrm{d}u = \left[2u - \frac{u^2}{2} + \frac{2u^{3/2}}{3} \right] \Big|_{1}^{2}$$

I will not complete the calculation here because the exact answer does not concern us, only the method. The number indicated above can be added to 4/3 to get the total area.

2.4 Taking Advantage of Symmetry

When a region has a line of symmetry, we can decompose it into two pieces that have equal area. We then only need to compute the area of one of the pieces.

Example 17. Consider the region between $x = 1 - y^2$ and $x = y^2 - 1$. [Draw picture] This has two lines of symmetry but for now we will consider just the y-axis. The total area is twice the area to the right of this, which can be computed as

$$\int_{-1}^{1} (1 - y^2) \mathrm{d}y = \left[y - \frac{y^3}{3} \right] \Big|_{-1}^{1} = 4/3$$

so the total area is 8/3.

Symmetry considerations can be applied to integrals more generally, even if we are not thinking of them as representing areas. In this case, the relevant symmetries are those of the function being integrated: whether it is even or odd.

Definition. Let f be a function.

1. f is said to be even if $f(-x) = f(x)$ for all x.
2. f is said to be odd if $f(-x) = -f(x)$ for all x.

Examples of even functions include even powers of x and the cosine function, while examples of odd functions include odd powers of x and the sine function.

Example 18. The function $f(x) = x^3 \sin(x) - x^2$ is even, because

$$f(-x) = (-x)^3 \sin(-x) - (-x)^2 = (-x^3)(-\sin(x)) - x^2 = x^3 \sin(x) - x^2 = f(x)$$

Proposition. *Consider a function f on a symmetric interval $[-a, a]$ around 0.*

1. *If f is odd, then*
$$\int_{-a}^{a} f(x)\mathrm{d}x = 0$$
2. *If f is even, then*
$$\int_{-a}^{a} f(x)\mathrm{d}x = 2\int_{0}^{a} f(x)\mathrm{d}x$$

This can be explained in two ways. One is geometric. [Draw pictures] For an odd function, the areas to the left and right of the y-axis cancel out, leaving zero. For even functions, the total area is the sum of two equal areas. The other way to explain this uses substitution. The point is that

$$\int_{-a}^{a} f(x)\mathrm{d}x = \int_{-a}^{0} f(x)\mathrm{d}x + \int_{0}^{a} f(x)\mathrm{d}x$$

Using the substitution $u = -x$, this is

$$\int_{a}^{0} -f(-u)\mathrm{d}u + \int_{0}^{a} f(x)\mathrm{d}x = \int_{0}^{a} f(-u)\mathrm{d}u + \int_{0}^{a} f(x)\mathrm{d}x = \int_{0}^{a} f(-x)\mathrm{d}x + \int_{0}^{a} f(x)\mathrm{d}x$$

For odd functions, these terms cancel out, while for even functions, they are equal.

Example 19. Consider the function $f(x) = \sin(2x^2)\sin(x^3)$. This satisfies

$$f(-x) = \sin(2(-x)^2)\sin((-x)^3) = \sin(2x^2)\sin(-x^3) = -\sin(2x^2)\sin(x^3) = -f(x)$$

so is odd. Therefore,

$$\int_{-1}^{1} f(x)\mathrm{d}x = 0$$

3 Problems for Group Work

Problem 1. Evaluate the indefinite integral

$$\int x \sin(2x^2 + 1)\sqrt{1 - \cos(2x^2 + 1)}\mathrm{d}x$$

Problem 2. Find the area of the region below $y = \cos(\pi x/2)$, above $y = -x - 2$, and to the left of $x = 1$. [Picture]